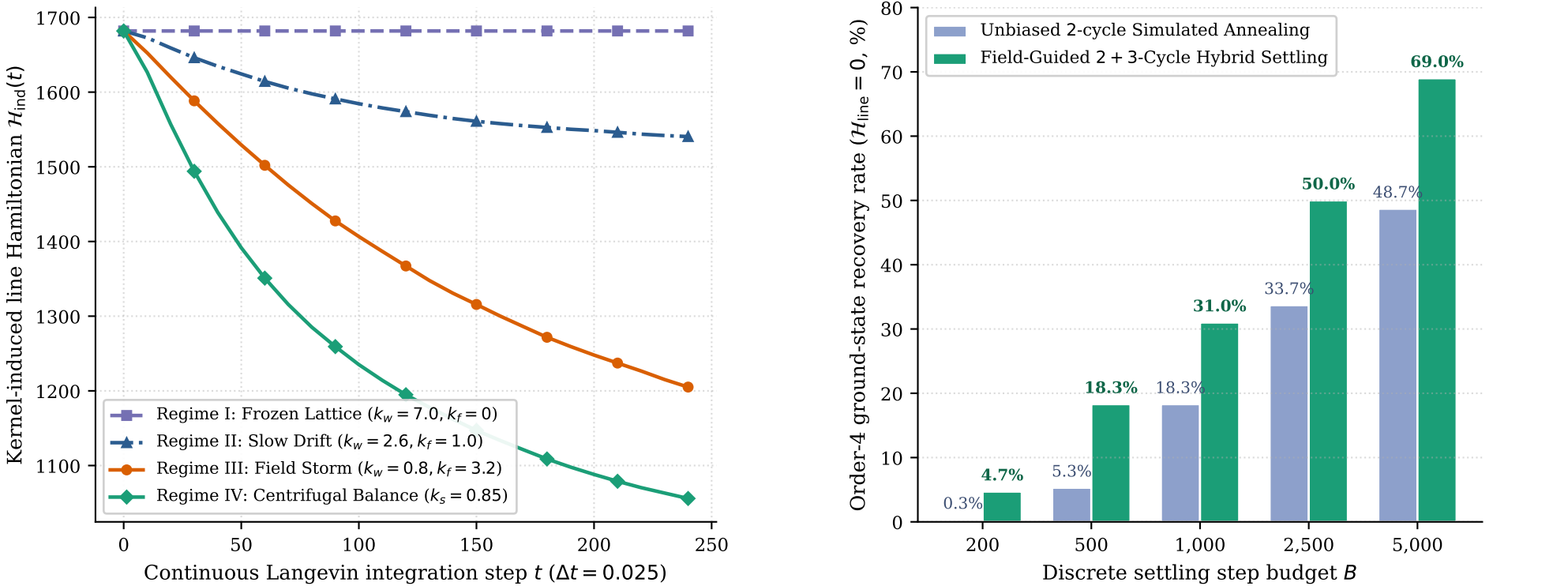
(a) Continuous Bubble-Trinity Relaxation (Regimes I--IV)
Kernel-induced line Hamiltonian $\mathcal{H}_{\mathrm{ind}}(t)$
Continuous Langevin integration step t ($\Delta t = 0.025$)
Regime I: Frozen Lattice ($k_w = 7.0, k_f = 0$)
Regime II: Slow Drift ($k_w = 2.6, k_f = 1.0$)
Regime III: Field Storm ($k_w = 0.8, k_f = 3.2$)
Regime IV: Centrifugal Balance ($k_s = 0.85$)
(b) Ground-State Synthesis: 2-Cycle SA vs. Hybrid Settling
Order-4 ground-state recovery rate ($\mathcal{H}_{\mathrm{line}} = 0$, %)
Discrete settling step budget B
Unbiased 2-cycle Simulated Annealing
Field-Guided 2 + 3-Cycle Hybrid Settling
0.3%
4.7%
5.3%
18.3%
18.3%
31.0%
33.7%
50.0%
48.7%
69.0%